HW3
Inverse Kinematics

Jacobian & CCD

Jacobian IK

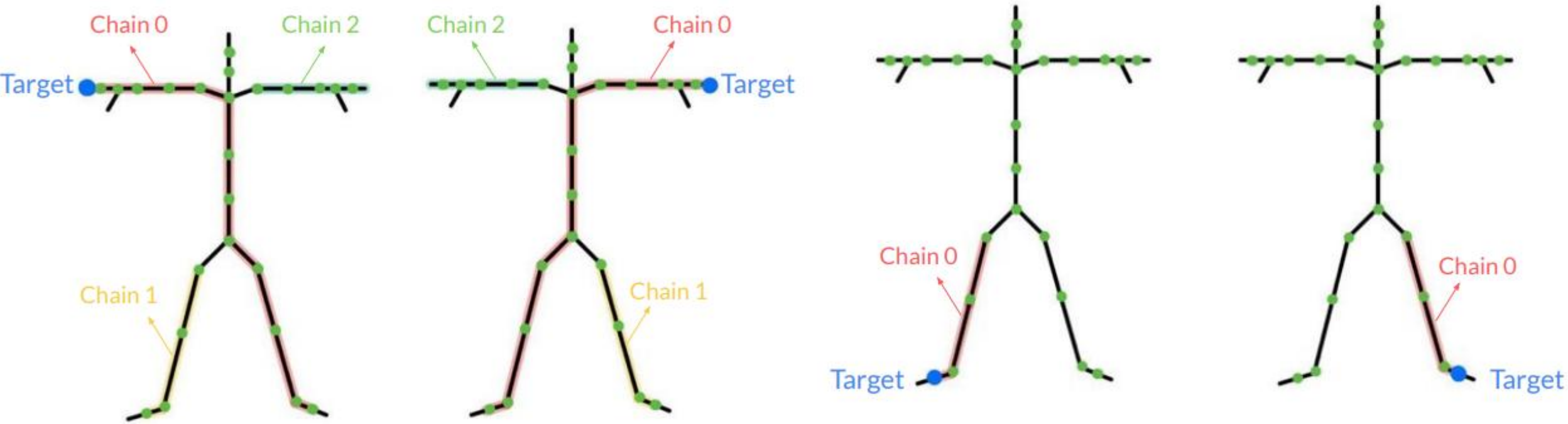
- JacobianChainSolver()
- pseudoInverseLinearSolver()

Review "kinematics.pptx" from p.20 - p.50

Review "acclaim_FK_IKnote.pdf" Inverse Kinematics part

Don’t Forget DOFs and rotation constraints.

Notabilia

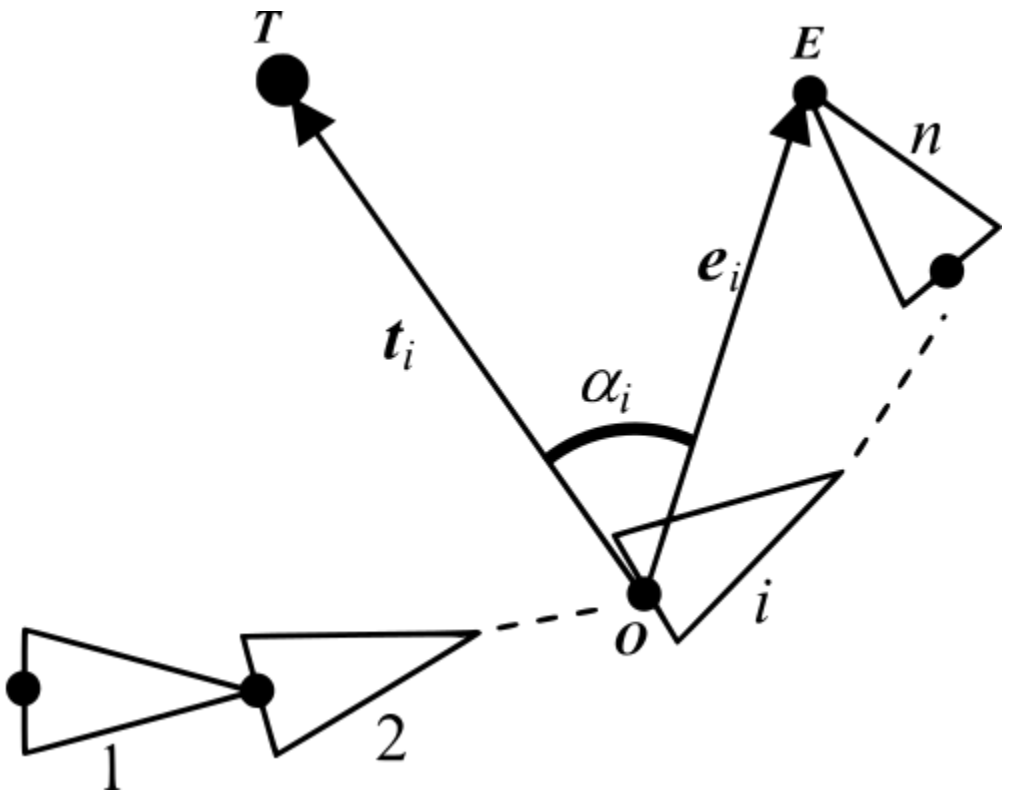

Chain 0
Chain 2
Target
Chain 1
Chain 2
Chain 0
Target
Chain 1
Chain 0
Target
Chain 0
Target

Notabilia

- Joint & Bone Chains



CCD IK: Overview



-Traverse Bone Chain

-Rotate Bone

-Update effector pos

CCD IK: Overview

Calculate rotation.

Convert world rotation to local.

Apply DOF & rotation constraints.

CCD IK: Overview

Updating effector position

Use the ACTUAL applied rotation.
(rotation was clamped)

Convert rotation back to world space.

Rotate vector to end effector.

Report

Question. 5%

Describe local and global coordinates in your words.

Implementation. 10%

Discussion. 5%

Different Step size.

If you noticed any difference between Jacobian and CCD.

Bonus. -

Optional, anything relevant to the topic.

Miscellaneous. -

Any changes outside of kinematics.cpp.

Other things you'd like to add.

Scoring

IKConverged()	5%
clampBoneRotation()	5%
pseudoInverseLinearSolver()	10%
JacobianChainSolver()	20%
CCDChainSolver()	20%
Jacobian Visual Correctness	10%
CCD Visual Correctness	10%
Report	20%
Bonus	up to 10%

Submission

HW3_<student_id>.zip

- kinematics.cpp
- report.pdf
- (any other modified files)

Deadline: 5/18 23:59

10% penalty / day after deadline.

If you have any questions please email all 3 TAs.